inclusion - exclusion $(|A \cup B| + |A \cap B| = |A| + |B|)$

ie $(rax \vee rdx) + (rax \wedge rdx) = rax + rdx$

ie $join(a,b) + meet(a,b) = a + b$

```
mov     rcx, rdx
and     rdx, rax
or      rax, rcx
add     rax, rdx
```

ie $z : L \longrightarrow \mathbb{N}$ is a "valuation" on a lattice

$\Updownarrow$

$z(a \vee b) + z(a \wedge b) = z(a) + z(b)$

useless, pedantic cat. :)

~~rdx = rax~~

$rcx = rdx$

$rdx = rdx_0 \wedge rax$

$rax = rax \vee rcx = rax_0 \vee rdx_0$

$rax = rax + rdx_0$
$= (rax_0 \vee rdx_0) + (rdx_0 \wedge rax_0)$
$= (rax_0 \vee rdx) + (rax \wedge rdx) = rax + rdx_0$